A Dynamical Analysis of Espresso Flow

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Introduction & Research Goal

Espresso extraction involves complex coupling between non-Newtonian fluid dynamics, porous media, and thermal evolution. While previous models [1][2] rely on simplified 1D Darcy approximations, we develop a general framework combining Navier-Stokes and Brinkman equations to account for anisotropic permeability and time-and-temperature-varying viscosity. By incorporating the permeability tensor and thermal transport, this theoretical framework provides analytical and numerical insights into flow uniformity for future experimental studies.

Analysis

Consider the fluid domain

$$\Omega_P = \Omega \setminus \bigcup_{i=1}^{N} B_\epsilon(p_i)$$

representing an ideal perforated cylindrical geometry, with a smooth mapping $\Phi : \Omega_P \longrightarrow \Omega_{\tilde{P}}$ that approximates the physical portafilter geometry.

We begin by recalling the incompressible Navier-Stokes equation,

$$\rho\left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v}\cdot\nabla\mathbf{v}\right) = -\nabla p + \nabla\cdot\boldsymbol{\sigma} + \mathbf{f}$$

where we may consider an ideal laminar vertical flow such that

$\mathrm{Re} = \frac{\rho v d}{\mu} \ll 1$ and obtain

$$\frac{\partial p}{\partial z} \approx \nabla\cdot\boldsymbol{\sigma}.$$

For a non-Newtonian fluid, the Cauchy stress tensor is given by

$$\boldsymbol{\sigma} = 2\mu(t,\dot{\gamma},T)\mathbf{D},$$

where

$$\mathbf{D} = \frac{1}{2}\left(\nabla\mathbf{v} + (\nabla\mathbf{v})^T\right).$$

To model this particular flow, we consider a modified power-law viscosity with time-and-temperature dependence given by

$$\mu(t,\dot{\gamma},T) = \dot{\gamma}^{n-1}(\mu_0 e^{-Kt} + \mu_M)e^{-b(T-T_0)},$$

where $\dot{\gamma}$ is the shear rate [4][5], T_0 is the reference temperature, μ_M, μ_0, b, and K are coefficients. Clearly,

$$\mu(0,T) = C_0\mu_i\dot{\gamma}^{n-1}e^{-bT}, \quad \lim_{t\to\infty}\mu(t,T) = C_0\mu_M\dot{\gamma}^{n-1}e^{-bT},$$

$$\mu(t,0) = C_0\dot{\gamma}^{n-1}(\mu_0 e^{-Kt} + \mu_M), \quad \lim_{T\to\infty}\mu(t,T) = 0,$$

where $\mu_i = \mu_0 + \mu_M$ and $C_0 = e^{bT_0}$.

By Darcy's law, the volumetric flow rate through the medium is given by

$$Q(z,t) = \frac{kA}{\mu(t,\dot{\gamma},T)}\frac{\Delta p}{\Delta h} = \frac{\pi R^2 k}{\mu(t,\dot{\gamma},T)}\frac{\partial p}{\partial z},$$

where k is the permeability of the medium. Using the shear rate of shear-thinning fluids in pipe flow,

$$\dot{\gamma} = \alpha\frac{v_z}{h},$$

we obtain the Darcy velocity

$$v_z = \frac{k}{\mu}\frac{\partial p}{\partial z} = \frac{ke^{bT}\,\partial_z p}{C_0(\alpha\frac{v_z}{h})^{n-1}(\mu_0 e^{-Kt} + \mu_M)},$$

which explicitly simplifies to

$$v_z = K_0\left(\frac{\partial_z p}{\mu_0 e^{-Kt} + \mu_M}\right)^{\frac{1}{n}} e^{\frac{bT}{n}},$$

where $K_0 = (\frac{h}{\alpha})^{\frac{n-1}{n}}(\frac{k}{C_0})^{\frac{1}{n}}$, in analogy with power-law porous-media flow [6]. This result also yields

$$Q \propto \left(\frac{\Delta p}{\Delta h}\right)^{\frac{1}{n}},$$

consistent with the volumetric flow relation [7].

One may recognize that

$$\nabla\cdot 2\mu(t,\dot{\gamma},T)\mathbf{D} = 2(\partial_j\mu)D_{ij} + 2\mu\partial_j D_{ij} \approx 2\mu\nabla\cdot\mathbf{D}$$

and

$$\nabla\cdot\mathbf{v} = 0, \quad \nabla\cdot\mathbf{D} = \frac{1}{2}\nabla^2\mathbf{v}$$

and find

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v_z}{\partial\theta^2} + \frac{\partial^2 v_z}{\partial z^2} + \frac{v_z}{k} = 0$$

under the assumption that the flow is approximately vertical. It can further be generalized into its vectorial form,

$$\nabla^2\mathbf{v} + \frac{1}{k}\mathbf{v} = 0,$$

the Darcy-Brinkman equation.

In fact, one of the main factors that contribute to a poor shot of espresso comes from anisotropy. To account for directional permeability, we consider the permeability tensor $\mathbf{K}$ and the new Darcy velocity [8],

$$\mathbf{v} = -\frac{1}{\mu}\mathbf{K}\cdot\nabla p,$$

For an ideal flow, the velocity is still approximately vertical, and its vertical component is now given by

$$v_z(r,\theta,z) = \frac{1}{\mu}\left(K_{zr}\partial_r p + K_{z\theta}\frac{1}{r}\partial_\theta p + K_{zz}\partial_z p\right)$$

and the total volumetric flow rate is found using

$$Q(t) = \int_{\partial_B\Omega_P}\mathbf{v}(r,\theta,0)\cdot d\mathbf{S} \approx \int_{\partial_B\Omega_P} v_z(r,\theta,0)\,dS.$$

The heat equation with viscous dissipation is given by

$$\rho c_p\frac{DT}{Dt} = \kappa\nabla^2 T + \Phi,$$

where ρ is the fluid density, c_p is the specific heat, κ is the thermal conductivity, and Φ is the viscous dissipation term,

$$\Phi = 2\mu D_{ij}D_{ij} = \mu\left[\left(\frac{\partial v_z}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial v_z}{\partial\theta}\right)^2 + 2\left(\frac{\partial v_z}{\partial z}\right)^2\right].$$

Moreover, invoking the rate of work identity

$$\Phi = \nabla\cdot(\boldsymbol{\sigma}\cdot\mathbf{v}) - \mathbf{v}\cdot\nabla p$$

where the first term vanishes upon integration due to the no-slip boundary condition and the divergence theorem, one may then find that the dissipation is given by the quadratic form

$$\Phi = -\mathbf{v}\cdot\nabla p = \frac{1}{\mu}(\mathbf{K}\cdot\nabla p)\cdot\nabla p = \frac{1}{\mu}(\nabla p)^T\mathbf{K}(\nabla p),$$

a positive-semidefinite quantity since $\mathbf{K} = \mathbf{K}^T$ thus ensuring non-negative entropy production in accordance with the second law of thermodynamics.

Figures and Results & Conclusion

We first define the following nondimensional parameters:

$$\tilde{t} = Kt, \quad \tilde{T} = b(T - T_0), \quad \tilde{\mu} = \mu/(\dot{\gamma}_0^{n-1}\mu_M), \quad \mu_N = \mu_0/\mu_M, \quad \tilde{\dot{\gamma}} = \dot{\gamma}/\dot{\gamma}_0,$$

and obtain the nondimensional power-law viscosity

$$\tilde{\mu} = \tilde{\dot{\gamma}}^{n-1}(1 + \mu_N e^{-\tilde{t}})e^{-\tilde{T}}.$$

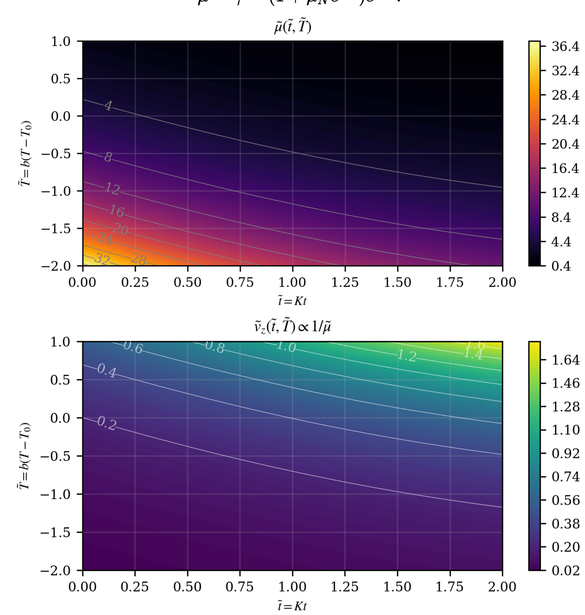

Fig. 1a & 1b. Contour map of the dimensionless viscosity and the dimensionless fluid velocity with time and temperature evolution, respectively.

To confirm that the proposed viscosity is consistent, we solve the Darcy-Brinkman equation imposing appropriate boundary and initial conditions [3] and obtain the separable solution

$$v_z(\mathbf{r},\tilde{t},\tilde{T}) = \left[\sum_{\lambda,n} J_\lambda(m_{\lambda,n}r)\Theta_\lambda(\theta)Z_n(z)\right]\cdot\frac{e^{\tilde{T}}}{1 + \mu_N e^{-\tilde{t}}},$$

$$\Theta_\lambda(\theta) = A\cos(\lambda\theta) + B\sin(\lambda\theta);\ \lambda\in\mathbb{Z},\ Z_n(z) = C_1 e^{-\beta_n z} + C_2 e^{\beta_n z},\ \beta\in\mathbb{R}.$$

We choose the fundamental mode $\mathrm{n} = 1$, $\lambda = 0$, the dominant mode for a pipe flow. Imposing this fundamental mode constraint reduces the solution from the Darcy-Brinkman PDE to a much simpler one, thus we obtain the nondimensional solution

$$\tilde{v}_{0,1}(\mathbf{r}_0,\tilde{t},\tilde{T}) \propto \frac{e^{\tilde{T}}}{1 + \mu_N e^{-\tilde{t}}},$$

which is precisely what is expected from the analysis of the proposed power-law viscosity.

We may define the thermal sensitivity coefficient b and the viscous resistance μ_N to characterize thermal thinning, mechanical resistance, and chemical volatility, as visualized in Fig, 2.

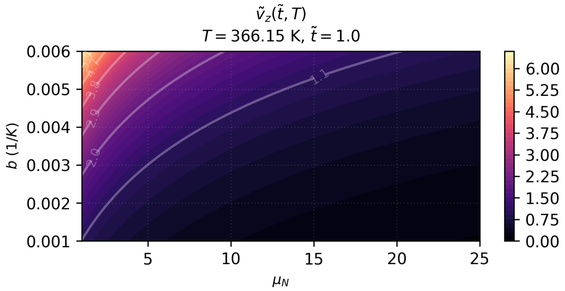

Fig. 2. Contour map of the dimensionless fluid velocity with varying thermal sensitivity coefficient b and viscous resistance, μ_N, at fixed time and temperature.

As a quantitative factor to measure uniformity, we define the normalized flux variance

$$S(r,\theta) = \frac{\sqrt{\int(Q(r,\theta,0) - \langle Q\rangle)^2\,dS}}{Q_t}$$

where the average flux is given by

$$\langle Q\rangle = \frac{1}{\pi R^2}\int_{\partial_B\Omega_P} Q(r,\theta)\,dS.$$

In numerical calculations, it can be realized that [3]

$$S = \frac{\sqrt{\frac{1}{N}\sum_{n=1}^{N}(Q_n - \langle Q\rangle)^2}}{\sum_{n=1}^{N} Q_n},$$

where Q_n is the n-th volumetric flow through each partitioned area.

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[8] Here, I have used the negative sign convention for the fluid velocity vector.